

For which Hamiltonians can we use the parameter-shift rule in VQE?

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Key point

The **parameter-shift rule** depends on the generators of the parametrized gates in the **ansatz**, *not* directly on the Hamiltonian.

- VQE objective:

$$E(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle, \quad |\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta}) |0\rangle.$$

- The Hamiltonian matters insofar as we must be able to **measure** $E(\boldsymbol{\theta})$ (typically via a Pauli decomposition).

Typical measurement form for H

In most applications (quantum chemistry, spin models, lattice gauge models),

$$H = \sum_k c_k P_k,$$

where each P_k is a tensor product of Pauli matrices (a “Pauli string”):

$$P_k \in \{I, X, Y, Z\}^{\otimes n}.$$

Then

$$E(\boldsymbol{\theta}) = \sum_k c_k \langle \psi(\boldsymbol{\theta}) | P_k | \psi(\boldsymbol{\theta}) \rangle,$$

and each $\langle P_k \rangle$ can be obtained from measurement statistics (after local basis changes if needed).

Ansatz structure and generators

Assume the ansatz is a product of parametrized unitaries,

$$U(\boldsymbol{\theta}) = U_L(\theta_L) \cdots U_2(\theta_2) U_1(\theta_1),$$

and each parameter enters through

$$U_j(\theta_j) = e^{-i\theta_j G_j},$$

with G_j Hermitian (the generator).

When does (simple) parameter-shift work?

When G_j has **only two distinct eigenvalues**.

Most common case in VQE

A very common choice is

$$G_j = \frac{1}{2}P, \quad P \in \{I, X, Y, Z\}^{\otimes n}.$$

Then the eigenvalues of G_j are $\pm\frac{1}{2}$, and the standard two-term shift rule holds:

$$\frac{\partial E}{\partial \theta_j} = \frac{1}{2} \left[E(\theta_j + \frac{\pi}{2}) - E(\theta_j - \frac{\pi}{2}) \right],$$

where $E(\theta_j \pm \frac{\pi}{2})$ means evaluating the VQE energy with all other parameters fixed and shifting only θ_j .

Formal theorem (two-eigenvalue generators)

Theorem (Two-point parameter-shift rule)

Let

$$E(\theta) = \langle \psi | U(\theta)^\dagger H U(\theta) | \psi \rangle, \quad U(\theta) = e^{-i\theta G},$$

where H and G are Hermitian and G has exactly two eigenvalues $\pm r$ (with $r > 0$). Then

$$\frac{dE}{d\theta} = r \left[E\left(\theta + \frac{\pi}{4r}\right) - E\left(\theta - \frac{\pi}{4r}\right) \right].$$

Equivalently,

$$\frac{dE}{d\theta} = \frac{1}{2s} [E(\theta + s) - E(\theta - s)], \quad s = \frac{\pi}{4r}.$$

Proof (spectral two-eigenvalue structure)

Proof.

Since G has eigenvalues $\pm r$, define the projectors onto the eigenspaces,

$$\Pi_{\pm} = \frac{1}{2} \left(I \pm \frac{G}{r} \right), \quad G = r(\Pi_+ - \Pi_-), \quad \Pi_{\pm}^2 = \Pi_{\pm}, \quad \Pi_+ \Pi_- = 0.$$

Then

$$U(\theta) = e^{-i\theta G} = e^{-i\theta r} \Pi_+ + e^{+i\theta r} \Pi_-.$$

Hence $E(\theta)$ is a trigonometric polynomial with frequency $2r$:

$$E(\theta) = A + B \cos(2r\theta) + C \sin(2r\theta),$$

for real constants A, B, C determined by $H, |\psi\rangle, \Pi_{\pm}$. Differentiate:

$$E'(\theta) = -2rB \sin(2r\theta) + 2rC \cos(2r\theta).$$

Interpretation and what depends on H

- The theorem does **not** require any special form of H beyond Hermiticity.
- What is required in practice is the ability to **estimate** $E(\theta)$ on hardware:

$$H = \sum_k c_k P_k \quad \Rightarrow \quad E(\theta) = \sum_k c_k \langle \psi(\theta) | P_k | \psi(\theta) \rangle .$$

- Therefore: **parameter-shift works for essentially any Hamiltonian used in VQE**, as long as you can measure its terms, *and* your ansatz gates have two-eigenvalue generators.

Examples of Hamiltonians used in VQE

Quantum chemistry (after mapping)

Fermionic Hamiltonians (second quantization) mapped via Jordan–Wigner / Bravyi–Kitaev become Pauli sums:

$$H = \sum_k c_k P_k.$$

Spin models

Ising:

$$H = \sum_i h_i Z_i + \sum_{ij} J_{ij} Z_i Z_j.$$

Heisenberg:

$$H = \sum_{ij} J_{ij} (X_i X_j + Y_i Y_j + Z_i Z_j).$$

All of these are directly Pauli decompositions $\Rightarrow$ measurable energy.

When the simple two-point rule fails

The simple two-evaluation shift rule can fail if the *generator* has more than two eigenvalues.

- Example (schematic): $U(\theta) = e^{-i\theta(X+Z)}$.
- Here $G = X + Z$ has a richer spectrum (more than two distinct eigenvalues for multi-qubit generalizations).

What then?

You may need:

- generalized multi-shift rules (more evaluations),
- linear-combination-of-unitaries constructions,
- or alternative gradient estimators (finite differences, SPSA, adjoint on simulators).

Answer to “for which Hamiltonians?”

For (almost) any Hamiltonian used in VQE, provided it can be measured (typically via a Pauli decomposition), *and* the ansatz gates are generated by operators with two distinct eigenvalues (e.g. Pauli strings).

- Hamiltonian requirement (measurement): $H = \sum_k c_k P_k$ (or can be reduced to this).
- Ansatz requirement (parameter-shift): each $U_j(\theta_j) = e^{-i\theta_j G_j}$ with $\text{spec}(G_j) = \{\pm r_j\}$.

One-slide summary

- Parameter-shift is controlled by the **gate generators** in the ansatz, not H .
- If each generator has two eigenvalues $\pm r$, then a **two-point shift rule** gives exact gradients.
- In practice VQE uses Hamiltonians decomposable into Pauli strings, so energies and shifted energies are measurable.